DANIEL IGBOKWE

Missoula, MT | danigbokwe@icloud.com | GitHub | LinkedIn

EDUCATION

Idaho State University — M.S. in Computer Science Conferred Aug 2025

Concentration: Data Analysis | GPA: 3.86/4.00 | Thesis defended July 2024

Thesis: Harnessing QUANtum Technology Integrated with Clustering for Sampling (QUANTICS) High Entropy Alloys (studied system: Ni-Co-Pt alloys)

Thesis advisor: Leslie Kerby

Idaho State University — B.S. in Computer Science Conferred Dec 2022

GPA: 3.55/4.00 | Cum laude

RESEARCH INTERESTS

Multimodal intelligence: perception, reasoning, memory, and action; human-in-the-loop AI; data-efficient machine learning.

PEER-REVIEWED PUBLICATIONS

K. Marchetti, A. Ševaljević, A. Diviney, C. Eardley, R. Phillips, R. Khadka, **D. Igbokwe**, and P. Bodily. "Redux: An Interactive, Dynamic Knowledge Base for Teaching NP-completeness." ITiCSE 2024, Vol. 1, pp. 255–261. ACM, 2024. DOI

A. Clark, **D. Igbokwe**, S. Ross, and M. F. Zibran. "A Quantitative Analysis of Quality and Consistency in AI-generated Code." 7th International Conference on Software and System Engineering (ICoSSE), pp. 37–41. IEEE, 2024. DOI

CONFERENCE ABSTRACTS

D. Igbokwe, L. Kerby, and M. Li. "Manifold-Guided Intelligent Sampling for Data-Efficient Screening of Ternary Ni-Co-Pt Alloys." 2027 MRS Spring Meeting & Exhibit, Symposium MT02. Abstract submitted October 14, 2026.

RESEARCH EXPERIENCE

Visual-Memory Reliability — Diagnostic Pilot Oct 2026

Personal research project | Offline RGB-D evaluation

- Built an auditable RGB-D replay and evaluation pipeline using DynaMem code; ran 120 object-location queries across two recorded DynaBench scenes: 54 development and 66 held out.
- Retrospectively compared age-only and age-plus-similarity ranking on 42 known-label held-out queries; the combined rule had 1 versus 2 geometric errors at 18 answers each, a tie-sensitive difference.
- Documented a wrong-object selection that passed a 30 cm location test; analyzed frame-cutoff sensitivity and excluded unknown labels from correctness scoring.

Graduate Research Assistant — QUANTICS Jan 2023 – Aug 2024

Center for Advanced Energy Studies, Idaho State University | Advisor: Leslie Kerby

- Developed a Python sampling workflow combining feature scaling, dimensionality reduction, clustering, and representative selection for hydrogen adsorption-energy modeling in the Ni-Co-Pt alloy system.
- Analyzed 5,400 candidate configurations with 12 features; compared random, PCA-guided, UMAP-guided, and literature-derived sampling using support-vector, kernel-ridge, XGBoost, and neural-network regressors.
- For HCP-site adsorption-energy prediction, XGBoost trained on 63 UMAP-selected configurations reached $R^2 = 0.980$ on 63 disjoint PCA-selected configurations, versus 0.956 for XGBoost trained on one random sample set (58 configurations after positive-energy exclusions).
- Extracted representative samples for collaborators' adsorption-energy calculations and used the returned labels for ML evaluation.
- Completed an M.S. thesis supported by INL's Laboratory Directed Research and Development program.

AI-Generated Code Quality Analysis 2024

Idaho State University | CS 6620 research project | Faculty supervisor: Minhaz F. Zibran

- Co-developed Python analysis scripts and statistical comparisons for a collaborative study of 625 Python samples and regenerated counterparts using seven Halstead measures.
- Co-authored the ICoSSE 2024 study, reporting broadly consistent code quality across iterative prompts and ChatGPT versions while emphasizing the need for code review.

Undergraduate Research Assistant — Redux Jan – Dec 2022

Computational Creativity and Intelligence Lab, Idaho State University | Supervisor: Paul Bodily

- Implemented Graph Coloring, including Karp's 3SAT-to-Graph-Coloring reduction, a solver, and a polynomial-time verifier; built interactive graph-instance and solution visualizations.
- Co-authored the ITiCSE 2024 paper. Redux's separate 13-participant usability study averaged 4.6/5 across 14 usability questions.

APPLIED AI AND SOFTWARE ENGINEERING

Software Developer — GCS Holdings, Inc. Dec 2024 – Present

- Built and deployed a Dockerized PDF OCR and document-extraction pipeline; processed approximately 3,000 PDFs end to end in 6–8 hours using OCR, local-LLM extraction, and deterministic parsing.
- Implemented page-type classification and submission of extracted well-history records to an ArcGIS layer, enabling users to view a well's history directly on the map.
- Implemented TimberVision AI's human-in-the-loop application, enabling field workers to review and correct drone-model tree detections within a system that automates plot assignment and reduces manual plot review.

Software Engineer Intern — Idaho State University Sep 2021 – May 2022

- Designed relational schemas, stored procedures, and validation triggers; documented software architecture and defects.

RESEARCH SOFTWARE

QUANTICS — Python Package for Intelligent Sampling 2024

Author and maintainer | PyPI: quantics

- Published a reusable implementation of normalization, dimensionality reduction, clustering, and representative-sample extraction used in the thesis workflow.

TEACHING EXPERIENCE

Teaching Assistant — Advanced AI Methods Spring 2024

Idaho State University | Instructor: Leslie Kerby

- Supported a graduate class of 20 students through office hours, assignment guidance, and explanations of difficult AI concepts.
- Graded assignments and midterms.

TECHNICAL SKILLS

Programming: Python, C#, JavaScript/TypeScript, SQL.

Machine learning and AI: scikit-learn, XGBoost, TensorFlow, pandas, NumPy; clustering, regression, experimental analysis; LLM/OCR inference (Ollama, Chandra OCR).

Engineering and deployment: Docker, Runpod, REST APIs, Git.

Geospatial and applications: ArcGIS JavaScript SDK, React, PostgreSQL, Microsoft SQL Server.